

PRACTICE TEST 07 - PCM

Time: 60 minutes Maximum marks: 64

PHYSICS

Section A (Single correct)

1. A ball is thrown vertically upward with a speed of 20 m/s. The maximum height reached is ($g = 10 \text{ m/s}^2$):
(A) 10 m (B) 20 m (C) 30 m (D) 40 m
2. A block of mass 2 kg rests on a rough horizontal surface with coefficient of friction 0.5. The minimum horizontal force needed to start moving it is ($g = 10 \text{ m/s}^2$):
(A) 5 N (B) 10 N (C) 15 N (D) 20 N
3. As shown in the figure, three resistors of 6 ohm each are connected between A and B. The equivalent resistance is:
(A) 2 ohm (B) 4 ohm (C) 9 ohm (D) 18 ohm
4. The work function of a metal is 2.3 eV. Light of photon energy 4.5 eV falls on it. The stopping potential is:
(A) 2.2 V (B) 6.8 V (C) 4.5 V (D) 2.3 V
6. A convex lens of focal length 20 cm forms an image of an object placed 40 cm from it. The image distance is:
(A) 20 cm (B) 40 cm (C) -40 cm (D) 80 cm

CHEMISTRY

Section A (Single correct)

1. The hybridisation of the central atom in SF_4 is:
(A) sp^3 (B) sp^3d (C) sp^3d^2 (D) dsp^2
2. A first-order reaction has a half-life of 20 min. The time for 75% completion is:
(A) 20 min (B) 30 min (C) 40 min (D) 60 min
3. Which of the following statements about a buffer solution are correct? (One or more options correct)
(A) It resists change in pH (B) It can be made from a weak acid and its salt (C) Its pH never changes (D) $\text{pH} = \text{pK}_a$ when $[\text{salt}] = [\text{acid}]$
4. The number of moles of O_2 required to completely burn 16 g of methane is:
(A) 1 (B) 2 (C) 3 (D) 4

MATHEMATICS

Section A (Single correct)

1. The value of $\lim_{x \rightarrow 0} \sin(5x)/x$ is:
(A) 0 (B) 1 (C) 5 (D) $1/5$
2. The length of the latus rectum of the parabola $y^2 = 12x$ is:
(A) 3 (B) 6 (C) 12 (D) 24
3. If $|A| = 3$ for a 3×3 matrix A, then $|\text{adj } A|$ equals:
(A) 3 (B) 9 (C) 27 (D) 81

Section B (Numerical value)

4. Find the area (in sq. units) bounded by $y = x^2$ and $y = 2x$.
5. Two dice are thrown. Find the probability that the sum is 7, correct to two decimal places.